

Eduardo Alonso / eddndev

SOFTWARE ENGINEER · ARCHITECTURE / PERFORMANCE /
SYSTEMS

Mexico City, MX

eddn.dev

achrony.me

github.com/eddndev

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• FOCUS

Engineer working at the intersection of **low-level systems**, **distributed computing** and **applied cryptography**. Designs language runtimes, bytecode VMs and zero-knowledge proof pipelines in Rust; ships entirely in-browser cryptographic tooling via WebAssembly. Comfortable owning a system end-to-end — from the constraint system up to the deploy pipeline.

• SELECTED PROJECTS

Achronyme

2024 — present · achrony.me

Functional language for zero-knowledge proofs. Compiles to three purpose-built bytecode VMs — Akron (40 opcodes), Artik (25), Lysis (29) — for constraint generation, witness computation and verification. Backends: Groth16, PlonK. Zero-trust pipeline with taint analysis and constraint-system verification.

[Rust](#) · [Custom VMs](#) · [Groth16](#) / [PlonK](#) · [Taint analysis](#)

Syle Studio

2024

Custom streetwear e-commerce on a proprietary CMS — no off-the-shelf platform. Owned schema design, performance budget and deploy pipeline.

[Node.js](#) · [PostgreSQL](#) · [Custom CMS](#)

crypto.eddn.dev

2026 ·

github.com/eddndev/crypto

Bilingual (EN/ES) educational cryptography suite: affine cipher, BMP steganography, matrix calculator, RSA hybrid encryption (AES-256-GCM + RSA PKCS#1 v1.5), AES modes of operation. Every algorithm is implemented in Rust, compiled to WASM, and executes 100% in the browser — keys and plaintext never leave the tab.

[Rust](#) · [WebAssembly](#) · [Astro](#) · [React islands](#) · [Tailwind v4](#)

Yatagarasu

2025

Zero-JavaScript portfolio site with retro-arcade aesthetic. Static rendering, no client runtime, sub-100ms interaction-to-paint targets.

[Astro](#) · [Zero-JS](#) · [Static rendering](#)

• ENGINEERING STACK

LANGUAGES

[Rust](#) (primary) · [TypeScript](#) · [Go](#) · [Dart](#) · [C](#) / [C++](#) · [Python](#)

SYSTEMS

[WebAssembly](#) · [Custom bytecode VMs](#) · [Compiler design](#) · [Memory-safe FFI](#)

CRYPTO

[Zero-knowledge proofs](#) (Groth16, PlonK) · [AES-GCM](#) · [RSA](#) · [PKCS#1](#) · [Constraint systems](#)

DISTRIBUTED

[Service architecture](#) · [Async runtimes](#) (Tokio) · [Message-passing](#) · [Containerized deploys](#)

WEB

[Astro](#) · [React](#) · [TypeScript](#) · [GSAP](#) · [Tailwind](#)

INFRA

[Docker](#) · [AWS](#) · [Cloudflare Pages](#) · [PostgreSQL](#) · [MongoDB](#)

• SELECTED ACHIEVEMENTS

- LeetCode **Guardian** — Contest Rating **2,749** (Top 0.05% globally); 160 Hard problems solved, 92.55% acceptance rate.
- Designed and shipped a three-VM bytecode pipeline for zero-knowledge proof generation, with separate VMs for constraint emission, witness solving and verification.
- Implemented a hybrid AES-256-GCM + RSA PKCS#1 v1.5 file-encryption scheme entirely in Rust → WASM, with all keying material confined to the browser tab.
- Built and operate eddndev — an independent digital engineering studio focused on custom infrastructure over template-based solutions.

• INTERESTS

Compiler internals

Zero-knowledge proofs

Applied cryptography

Distributed systems

WebAssembly

Competitive programming

VM design